

STOCHASTIC VOLATILITY AND JUMP

Sergey Mazyavkin[†]

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1 Partial Differential Equation

Suppose the underlying asset value follows the diffusion:

$$dS_t = \alpha S_t dt + \sqrt{v_t} S_t dW_t^{S, \mathbb{P}} + (Y_t - 1) S_t dN_t^{\mathbb{P}}$$

where α is the mean growth rate, and $W_t^{\mathbb{P}}$ is the Brownian motion under the physical probability measure, $N_t^{\mathbb{P}}$ is a Poisson jump process with intensity $\bar{\gamma}$ under measure $\mathbb{P}$. When a jump occurs ($dN_t^{\mathbb{P}} = 1$), the asset price is multiplied by a random jump factor $Y_t > 0$ (so $S_t = Y_t S_{t-}$ at the jump time). In fact, $Y_t = e^{J_t}$, so that J_t is the jump size in log-price. I assume J_t are i.i.d. with distribution under $\mathbb{P}$ characterised by mean $\bar{\mu}_J$ and variance σ_J^2 ¹. Importantly, the total (continuous) growth rate in S_t is $\mu - \bar{\kappa} = \alpha + \delta$, where $\bar{\kappa}$ is the mean relative jump size $\mathbb{E}^{\mathbb{P}}[Y_t - 1]$ ², and δ is the return shortfall (the cashflow the underlying asset requires or offers for holding an asset, e.g. the dividend yield for stocks).

Further, v_t is the variance following Cox-Ingersoll-Ross (1985) square-root process:

$$dv_t = \kappa(\bar{v} - v_t)dt + \xi\sqrt{v_t}dW_t^{v, \mathbb{P}}$$

with κ being the speed of mean reversion, $\bar{v}$ long-run average variance, and ξ is the volatility of variance parameter. $W_t^{S, \mathbb{P}}$ and $W_t^{v, \mathbb{P}}$ are two brownian motions driving innovations, such that $\mathbb{E}^{\mathbb{P}}[dW_t^{S, \mathbb{P}} dW_t^{v, \mathbb{P}}] = \rho dt$,

[†]Accounting and Finance Division, Booth Street West, Manchester M15 6PB, UK, tel.: +44(0)161 306 0322, fax: +44(0)161 306 0322, e-mail: sergey.mazyavkin@manchester.ac.uk. All errors are my own.

¹For example, $J_t \sim \mathcal{N}(\bar{\mu}_J, \sigma_J^2)$ for log-normal jumps.

² $e^{\bar{\mu}_J + \frac{1}{2}\sigma_J^2} - 1$ when $J_t \sim \mathcal{N}(\bar{\mu}_J, \sigma_J^2)$, so $\bar{\kappa}$ accounts for the average percentage jump. It ensures that $\mathbb{E}^{\mathbb{P}}[dS_t/S_t] = \mu dt$.

with ρ , normally, being less than zero³, while $N_t^{\mathbb{P}}$ is independent of Brownian motions. Intuitively, κ controls the persistence of variance, ρ and ξ shape the distribution's skewness and kurtosis via the diffusion channel, while the jump parameters – intensity γ , mean size μ_J , and volatility σ_J – introduce additional negative skewness and fat tails⁴.

Note that with two sources of randomness ($W_t^{S,\mathbb{P}}$ and $W_t^{v,\mathbb{P}}$) and additional jump risk, the market is incomplete – not all risks can be perfectly hedged by trading only in the underlying asset and a risk-free bond. This means we cannot replicate all contingent payoffs with a self-financing portfolio. As a result, option pricing will depend on risk premia for variance risk and jump risk. In what follows, I incorporate parameters for these risk premia when moving to an arbitrage-free (risk-neutral) pricing measure. In particular, I apply extended (to jump-diffusion processes) Feynman-Kac theorem to the system of SDEs (11) & (10):

$$\begin{aligned} \frac{\partial V}{\partial t} + (r - \delta - \gamma\kappa)S_t \frac{\partial V}{\partial S} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial V}{\partial v} + \frac{1}{2}S_t^2 v_t \frac{\partial^2 V}{\partial S^2} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 V}{\partial v^2} + \rho\xi v_t S_t \frac{\partial^2 V}{\partial S \partial v} \\ + \gamma \mathbb{E}^{\mathbb{Q}}[V(S_t Y_t, v_t, t) - V(S_t, v_t, t)] = rV \end{aligned} \quad (1)$$

which is exactly an option pricing PDE⁵ – existence of the risk-neutral measure $\mathbb{Q}$ ensures that there is no free lunch (first fundamental theorem of asset pricing). The PDE underpins the corresponding Hamilton-Jacobi-Bellman equation if I were to find an option price using the dynamic programming.

2 European Call Option

I first, with the help of Itô's Lemma, restate the Equation (1) in terms of $s = \ln S$ ⁶:

$$\begin{aligned} \frac{\partial V}{\partial t} + \left(r - \delta - \gamma\mu_J - \frac{1}{2}v_t\right) \frac{\partial V}{\partial s} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial V}{\partial v} + \frac{1}{2}v_t \frac{\partial^2 V}{\partial s^2} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 V}{\partial v^2} + \rho\xi v_t \frac{\partial^2 V}{\partial s \partial v} \\ + \gamma \mathbb{E}^{\mathbb{Q}}[V(s_t + J_t, v_t, t) - V(s_t, v_t, t)] = rV \end{aligned} \quad (2)$$

so that European call option becomes

$$\mathcal{C}(K) = e^{-r\tau} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] \equiv e^{-\delta\tau} e^s P_1 - e^{-r\tau} K P_2 \quad (3)$$

³See, e.g. Black (1976) for a note on so-called “leverage effect.”

⁴Setting $\kappa = \xi = \gamma = 0$ recovers the Black and Scholes (1973) model.

⁵Consider, e.g. European put option with the terminal condition $V(S_T, v_T, T) = (K - S_T)^+$.

⁶To get here quick, apply Itô's Lemma to get to the SDE (12), then apply Feynman-Kac theorem described in the Section 5 to the SDE system with $\ell = (s_t - \ln K)^+$.

where K is the strike price, $\tau = T - t$ is the time to maturity, P_1 and P_2 are exercise probabilities such that⁷

$$P_1 = \mathbb{Q}'\{S_T > K\} \quad (4)$$

$$P_2 = \mathbb{Q}\{S_T > K\} \quad (5)$$

where $\mathbb{Q}' \sim \mathbb{Q}$ is the equivalent martingale measure that uses $S_t e^{\delta t}$ as numéraire. I recognise that P_1 and P_2 are the exercise probabilities and thus must satisfy their respective PDEs – and apply Feynman-Kac theorem. Since $P_2 = \mathbb{E}^{\mathbb{Q}}[\mathbb{I}_{\{S_T > K\}}]$ ⁸, P_2 must satisfy the PDE (15) with $r(x, t) = 0$ (intuitively, pure probability is not discounted, it must be bounded between 0 and 1 at any point in time) and $f(x, t) = 0$, therefore,

$$\begin{aligned} \frac{\partial P_2}{\partial t} + \left(r - \delta - \gamma\mu_J - \frac{1}{2}v_t\right) \frac{\partial P_2}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 P_2}{\partial s^2} + (\kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial P_2}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 P_2}{\partial v^2} + \xi \rho v_t \frac{\partial^2 P_2}{\partial s \partial v} \\ + \gamma \mathbb{E}^{\mathbb{Q}}[P_2(s_t + J_t, v_t, t) - P_2(s_t, v_t, t)] = 0 \end{aligned}$$

As for the P_1 , I go in the exact same manner. However, from Bakshi and Madan (2000) representation in the Equations (4), $P_1 = \mathbb{E}^{\mathbb{Q}'}[\mathbb{I}_{\{S_T > K\}}]$. I, therefore, first obtain the system of SDEs (13) and (14) under $\mathbb{Q}'$. Then P_1 must satisfy the PDE (15) with $r(x, t) = 0$ (again, pure probability is not discounted) and $f(x, t) = 0$:

$$\begin{aligned} \frac{\partial P_1}{\partial t} + \left(r - \delta - \gamma\mu_J + \frac{1}{2}v_t\right) \frac{\partial P_1}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 P_1}{\partial s^2} + (\rho\xi v_t + \kappa(\bar{v} - v_t) - \lambda v_t) \frac{\partial P_1}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 P_1}{\partial v^2} + \xi \rho v_t \frac{\partial^2 P_1}{\partial s \partial v} \\ + \gamma \mathbb{E}^{\mathbb{Q}'}[P_1(s_t + J_t, v_t, t) - P_1(s_t, v_t, t)] = 0 \end{aligned}$$

I can state the probabilities is “Heston” form:

$$\begin{aligned} \frac{\partial P_j}{\partial t} + (r - \delta - \gamma\mu_J - u_j v_t) \frac{\partial P_j}{\partial s} + \frac{1}{2}v_t \frac{\partial^2 P_j}{\partial s^2} + (a - b_j v_t) \frac{\partial P_j}{\partial v} + \frac{1}{2}\xi^2 v_t \frac{\partial^2 P_j}{\partial v^2} + \xi \rho v_t \frac{\partial^2 P_j}{\partial s \partial v} \\ + \gamma \mathbb{E}^{\mathbb{Q}^j}[P_j(s_t + J_t, v_t, t) - P_j(s_t, v_t, t)] = 0 \end{aligned} \quad (6)$$

where $j \in (1, 2)$ $u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $a = \kappa\bar{v}$, $b_1 = \kappa + \lambda - \rho\xi$, $b_2 = \kappa + \lambda$, $\mathbb{Q}_1 = \mathbb{Q}'$, and $\mathbb{Q}_2 = \mathbb{Q}$.

Gil-Pelaez (1951) derives the expression for the probability in terms of the characteristic function of s_T under $\mathbb{Q}_j$, $\varphi_j(\cdot)$:

$$\mathbb{P}\{s_T > k\} \equiv P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iuk} \varphi_j(u)}{iu} \right] du \quad (7)$$

⁷This is Bakshi and Madan (2000) representation of call option. The representation works for a wide range of models, so I show it here. For instance, those will be exactly $\mathcal{N}(d_1)$ and $\mathcal{N}(d_2)$ from the Black and Scholes (1973).

⁸i.e. $\ell = \mathbb{I}_{\{S_T > K\}}$

which is obtained via Fourier transform

$$P\{s_T > k\} = 1 - F_s(k) \equiv \int_k^\infty f_s(x)dx = \frac{1}{2\pi} \int_k^\infty \left(\int_{-\infty}^\infty e^{-iux} \varphi(u) du \right) dx \equiv \frac{1}{2\pi} \int_{-\infty}^\infty \varphi(u) \left(\int_k^\infty e^{-iux} dx \right) du$$

where $F_s(\cdot)$ ($f_s(\cdot)$) is the cdf (pdf) of s_T . I refer reader to a very elegant single-page solution resulting in (7) in Gil-Pelaez (1951). Intuitively, however, it is just as if we have the pdf and cdf.

3 Characteristic Function

Clearly, the distribution of s_T does not admit the pdf as it does in the case of Black and Scholes (1973). The characteristic function is, however, readily available. Having characteristic function, I am able to extract pdf with the Fourier transform. As a consequence of Feynman-Kac theorem, characteristic functions $\varphi_j(\phi, s_t, v_t)$ will follow their respective PDEs (6). The characteristic function *ansatz* is given in Duffie, Pan, and Singleton (2000)⁹:

$$\varphi_j(\phi, s_t, v_t) \equiv \mathbb{E}^{\mathbb{Q}_j}[e^{i\phi s_t}] = e^{C_j(\tau) + D_j(\tau)v_t + i\phi s_t}$$

which I substitute (with a little change $\tau = T - t$) in the PDE to get

$$\begin{aligned} -\frac{\partial \varphi_j}{\partial \tau} + (r - \delta - \gamma\mu_J - u_j v_t) \frac{\partial \varphi_j}{\partial s} + \frac{1}{2} v_t \frac{\partial^2 \varphi_j}{\partial s^2} + (a - b_j v_t) \frac{\partial \varphi_j}{\partial v} + \frac{1}{2} \xi^2 v_t \frac{\partial^2 \varphi_j}{\partial v^2} + \xi \rho v_t \frac{\partial^2 \varphi_j}{\partial s \partial v} \\ + \gamma \mathbb{E}^{\mathbb{Q}_j}[\varphi_j(\phi, s_t + J_t, v_t) - \varphi_j(\phi, s_t, v_t)] = 0 \end{aligned}$$

The compensator term collapses to

$$\begin{aligned} \mathbb{E}^{\mathbb{Q}_j}[\varphi_j(\phi, s_t + J_t, v_t) - \varphi_j(\phi, s_t, v_t)] &\equiv \int_{\mathbb{R}} [e^{C_j(\tau) + D_j(\tau)v_t + i\phi(s_t + J_t)} - e^{C_j(\tau) + D_j(\tau)v_t + i\phi s_t}] \phi(J) dJ \\ &= \int_{\mathbb{R}} \varphi_j(\phi, s_t, v_t) (e^{i\phi J_t} - 1) \phi(J) dJ \\ &= \varphi_j(\phi, s_t + J_t, v_t) (\mathbb{E}^{\mathbb{Q}_j}[e^{i\phi J_t}] - 1) \end{aligned}$$

where $\varphi_j(\phi, J) = \mathbb{E}^{\mathbb{Q}_j}[e^{i\phi J_t}]$ is the characteristic function of J_t . Also note that $\mathbb{E}^{\mathbb{Q}}[e^{i\phi J_t}] = \mathbb{E}^{\mathbb{Q}'}[e^{i\phi J_t}]$, so that J_t has the same distribution under both $\mathbb{Q}$ and $\mathbb{Q}'$, so I drop the subscript for characteristic function of J_t , $\varphi(\phi, J)$. So that substituting the characteristic function ansatz in the PDE yield the system of ODEs (Duffie,

⁹In fact, if there is another “affine” source of uncertainty, e.g. risk-free rate r_t , the characteristic function will take the form $\varphi_j(\phi, s_t, v_t) \equiv \mathbb{E}^{\mathbb{Q}_j}[e^{i\phi s_t}] = e^{C_j(\tau) + D_j(\tau)v_t + E_j(\tau)r_t + i\phi s_t}$.

Pan, and Singleton (2000) show that under general regularity conditions, the conditional characteristic function of affine jump-diffusions has the exponential-affine form with the coefficients being the ODE solutions):

$$\begin{cases} \frac{\partial D_j(\tau)}{\partial \tau} = \rho \xi i \phi D_j(\tau) - \frac{1}{2} \phi^2 + \frac{1}{2} \xi^2 D_j^2(\tau) + u_j i \phi - b_j D_j(\tau) \\ \frac{\partial C_j(\tau)}{\partial \tau} = (r - \delta - \gamma \mu_J) i \phi + a D_j(\tau) + \gamma (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) \end{cases}$$

where the last term appear from the normality assumption $J_t \sim \mathcal{N}(\mu_J, \sigma_J^2)$. The first ODE in $D_j(\tau)$ is a standard Heston's Riccati¹⁰ equation with the known solution, so I avoid deriving it here:

$$\begin{aligned} D_j(\tau) &= \frac{b_j - \rho \xi i \phi - d_j}{\xi^2} \frac{1 - e^{-d_j \tau}}{1 - c_j e^{-d_j \tau}} \\ d_j &= \sqrt{(b_j - \rho \xi i \phi)^2 - \xi^2 (2u_j i \phi - \phi^2)} \\ c_j &= \frac{b_j - \rho \xi i \phi - d_j}{b_j - \rho \xi i \phi + d_j} \end{aligned} \tag{8}$$

I further obtain $C_j(\tau)$ by integration:

$$\begin{aligned} C_j(\tau, \phi) &= \int_0^\tau (r - \delta - \gamma \mu_J) i \phi du + a v \frac{b_j - \rho \xi i \phi - d_j}{\xi^2} \int_0^\tau \frac{1 - e^{-d_j u}}{1 - c_j e^{-d_j u}} du + \gamma \int_0^\tau (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) du \\ &= (r - \delta - \gamma \mu_J) i \phi \tau + \gamma (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) \tau + a \frac{b_j - \rho \xi i \phi - d_j}{\xi^2} \int_0^\tau \frac{1 - e^{-d_j u}}{1 - c_j e^{-d_j u}} \frac{d_j e^{d_j u}}{d_j e^{d_j u}} du \\ &= (r - \delta - \gamma \mu_J) i \phi \tau + \gamma (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) \tau + \frac{a}{d_j} \frac{b_j - \rho \xi i \phi - d_j}{\xi^2} \int_1^{e^{d_j \tau}} \frac{1 - e^{-d_j u}}{(1 - c_1 e^{-d_j u}) e^{-d_j u}} d(e^{-d_j u}) \\ &= (r - \delta - \gamma \mu_J) i \phi \tau + \gamma (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) \tau + \frac{a}{d_j} \frac{b_j - \rho \xi i \phi - d_j}{\xi^2} \int_1^{e^{d_j \tau}} \left[\frac{1}{e^{d_j u}} - \frac{1 - e^{-d_j u}}{1 - c_j e^{-d_j u}} \right] d(e^{-d_j u}) \\ &= (r - \delta - \gamma \mu_J) i \phi \tau + \gamma (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) \tau + \frac{\kappa \bar{v}}{d_1} \frac{b_1 - \rho \xi i \phi - d_1}{\xi^2} \left[d_1 u - \frac{1 - c_1}{c_1} \ln(1 - c_1 e^{d_1 u}) \right]_{u=1}^{u=e^{d_1 \tau}} \\ &= (r - \delta - \gamma \mu_J) i \phi \tau + \gamma (e^{i \phi \mu_J - \frac{1}{2} \phi^2 \sigma_J^2} - 1) \tau + \frac{a}{\xi^2} \left((b_j - \rho \xi i \phi - d_j) - \ln \frac{1 - c_j e^{-d_j \tau}}{1 - c_j} \right) \end{aligned} \tag{9}$$

4 Equivalent Martingale Measure

To price options, I seek an equivalent martingale measure $\mathbb{Q}$ under which the discounted asset price is a martingale. Let r be the constant risk-free interest rate. We assume $\mathbb{Q}$ exists (by the first fundamental theorem of asset pricing) such that when discounting at r , all asset prices have no drift under $\mathbb{Q}$. In particular, the

¹⁰In fact, if there is another ‘‘affine’’ source of uncertainty, e.g. risk-free rate r_t , the system will contain three ODEs, the additional one will take a Riccati form as well, subject to parameters (in case r_t follows CIR process, the ODE will be exactly the same as for $D_j(\tau)$ with just ρ_r , ξ_r , κ_r , and λ_r).

asset's expected drift under $\mathbb{Q}$ must equal $r - \delta$ after accounting for jumps. This will determine the risk-neutral drift adjustments.

4.1 $\mathbb{Q}$ -Measure

I first follow Heston (1993) in adjusting the variance process

$$dv_t = (\kappa(\bar{v} - v_t) - \lambda v_t)dt + \xi\sqrt{v_t}dW_t^{v,\mathbb{Q}} \quad (10)$$

where λ is the variance risk premium parameter¹¹. Further, I assume that under the measure $\mathbb{Q}$, the jump intensity is γ and $J_t \sim \mathcal{N}(\mu_J, \sigma_J^2)$. So that, $\varkappa = \mathbb{E}^{\mathbb{Q}}[Y_t - 1] = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1$. Apparently, I also switch drift α to r ¹². The idea, as usual, is to have discounted S_t martingale under $\mathbb{Q}$:

$$dS_t = (r - \delta - \gamma\varkappa)S_t dt + \sqrt{v_t}S_t dW_t^{S,\mathbb{Q}} + (Y_t - 1)S_t dN_t^{\mathbb{Q}} \quad (11)$$

where $N_t^{\mathbb{Q}}$ is the Poisson process with intensity γ under $\mathbb{Q}$. Importantly, I do not change σ_J^2 parameter when switch from $\mathbb{P}$ to $\mathbb{Q}$. Although it would make economic sense – investors do not like jump uncertainty, so it would be reasonable to have $\sigma_J^2 < \bar{\sigma}_J^2$. However, since having jump already introduces extra skewness (via $\mu_J < 0$) and excess kurtosis, I avoid having an extra parameter.

It may further be convenient to establish the SDE in term of $s_t = \ln S_t$:

$$ds_t = \left(r - \delta - \gamma\mu_J - \frac{1}{2}v_t \right) dt + \sqrt{v_t}dW_t^{S,\mathbb{Q}} + J_t dN_t^{\mathbb{Q}} \quad (12)$$

which is obtained via Itô's Lemma¹³. A note on how we went from $\gamma\varkappa$ to $\gamma\mu_J$ is important. Note that $\mathbb{E}^{\mathbb{Q}}[dS_t] = (r - \delta)S_t dt$ with continuous part drift being $\mathbb{E}^{\mathbb{Q}}[dS_t] - \mathbb{E}^{\mathbb{Q}}[(Y_t - 1)dN_t] = (r - \delta - \gamma\varkappa)S_t dt$. Similarly, $\mathbb{E}^{\mathbb{Q}}[dS_t] - \mathbb{E}^{\mathbb{Q}}[J_t dN_t] = (r - \delta - \gamma\mu_J)S_t dt$.

¹¹By Girsanov's theorem, $dW_t^{v,\mathbb{Q}} = dW_t^{v,\mathbb{P}} + \frac{\lambda v_t}{\xi\sqrt{v_t}}dt$.

¹²Recall that the total continuous drift under $\mathbb{P}$ is $\mu - \bar{\varkappa}$. So that the total continuous drift under $\mathbb{Q}$ becomes $r - \delta - \varkappa$. Then apply Girsanov's theorem accordingly.

¹³For the $X_t = (X_t - X_{t-})dN_t$, $df(X_t) = (f(X_t) - f(X_{t-}))dN_t$ (you can expand more in Taylor series, yet papers (like Bates) do not do it. Effectively, they state the process for s_t that way, only with J_t . With $x_t = \ln X_t$, $ds_t = \ln \frac{X_t}{X_{t-}}$.

4.2 $\mathbb{Q}'$ – Measure

Next, I apply the change in numéraire so that to express the processes under the measure $\mathbb{Q}' \sim \mathbb{Q}$, the equivalent martingale measure that uses $S_t e^{\delta t}$ as numéraire. The change in numéraire is

$$\frac{d\mathbb{Q}'}{d\mathbb{Q}}|_t = \frac{S_T e^{\delta T} / S_t e^{\delta t}}{e^{rT} / e^{rt}}$$

such that

$$\mathbb{E}^{\mathbb{Q}}[S_T \mathbb{I}_{\{S_T > K\}}] = S_t e^{(r-\delta)\tau} \mathbb{Q}'\{S_T > K\}$$

and the resulting system of SDEs is

$$ds_t = \left(r - \delta - \gamma\mu_J + \frac{1}{2}v_t \right) dt + \sqrt{v_t} dW_t^{S, \mathbb{Q}'} \quad (13)$$

$$dv_t = (\rho\xi v_t + \kappa(\bar{v} - v_t) - \lambda v_t) dt + \xi \sqrt{v_t} dW_t^{v, \mathbb{Q}'} \quad (14)$$

5 Feynman-Kac Theorem

Here I extend Feynman-Kac theorem to accommodate for jumps. The Feynman-Kac theorem states that if we have the PDE of the form

$$\frac{\partial V}{\partial t} + \sum_{i=1}^N \mu_i(x, t) \frac{\partial V}{\partial x_i} + \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \sigma_{i,j}(x, t) \frac{\partial^2 V}{\partial x_i \partial x_j} - r(x, t)V + \gamma \mathbb{E}^{\mathbb{Q}}[V(x, t) - V(x, t^-)] = f(x, t) \quad (15)$$

with the terminal condition $V(x, T) = \ell(x)$, and the process $X_t \in \mathbb{R}^N$ satisfies

$$dX_t = \mu(X_t, t)dt + \Sigma(X_t, t)dW_t^{\mathbb{Q}} + (Y_t - 1)dN_t^{\mathbb{Q}}$$

then the solution is given by

$$V(x, t) = \mathbb{E}^{\mathbb{Q}} \left[\int_t^T e^{-\int_t^T r(X_\tau, \tau) d\tau} f(X_u, u) du + e^{-\int_t^T r(X_\tau, \tau) d\tau} \ell(X_T) \right]$$

so that in case of stochastic volatility plus jump model $X_t = \begin{bmatrix} S_t \\ v_t \end{bmatrix}$, $\mu_t = \begin{bmatrix} (r - q - \gamma\kappa)S_t \\ \kappa((\bar{v}_t - v_t) - \lambda v_t) \end{bmatrix}$, and $\Sigma_t = \sqrt{v_t} \begin{bmatrix} \rho S_t & \sqrt{1 - \rho^2} S_t \\ \xi & 0 \end{bmatrix}$, assuming $dW_t^{\mathbb{Q}} = \begin{bmatrix} dW_{1,t}^{\mathbb{Q}} \\ dW_{2,t}^{\mathbb{Q}} \end{bmatrix}$. Also, $r(x, t) = r$, $h(x, t) = 0$, $\ell(x) = (S_T - K)^+$, and, generally,

$$\mathbb{E}^{\mathbb{Q}}[V(x, t) - V(x, t^-)] = \int_{\mathbb{R}} [V(S_t e^J, v_t, t) - V(S_t, v_t, t)] \phi_J dJ$$

where $\phi(j)$ is the density of J_t .